

# Gene Expression Modelling Pipeline

## Modelling Gene Expression Using Motif Occurrence in the Enhancer Region in the Promoter Window of 150kb and 2kb and in the Active Regions Having H2BK20ac Histone Modification in GM12878 cell line

*Modeling Gene Expression from Motif Occurrences at Enhancers and Promoters*

Cell Line: GM12878 | Genome: hg38 (GRCh38.110) | Motif DB: JASPAR 2024

Group 12 | Data Science in Genomics | 2024

Ashwini Gudekar (MT25177)

Ishika Gupta(MT25280)

Sunandini Chowdhury(MT25187)

Pradip N Palekar(MT25215)

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# 1. Project Overview

This pipeline models gene expression (FPKM) using transcription factor (TF) motif occurrence scores at enhancers and promoters in the GM12878 human lymphoblastoid cell line. The core idea is to ask: *can the binding landscape of transcription factors near a gene predict how strongly that gene is expressed?* The final output is a gene x motif matrix linked to GM12878 FPKM values, ready for regression and machine learning modelling.

| Parameter       | Value                                                  |
|-----------------|--------------------------------------------------------|
| Cell line       | GM12878 (128 single-cell samples)                      |
| Genome assembly | hg38 / GRCh38.110                                      |
| Motif database  | JASPAR 2024 (~870 motifs)                              |
| Enhancers       | 65,125 peaks (enhancers.bed)                           |
| Promoter window | -50 kb to +100 kb from TSS (strand-aware)              |
| Final matrix    | 20,399 genes x 817 motifs (45.7% dense)                |
| FPKM source     | GSE81861_Cell_Line_FPKM.csv (GM12878 columns averaged) |

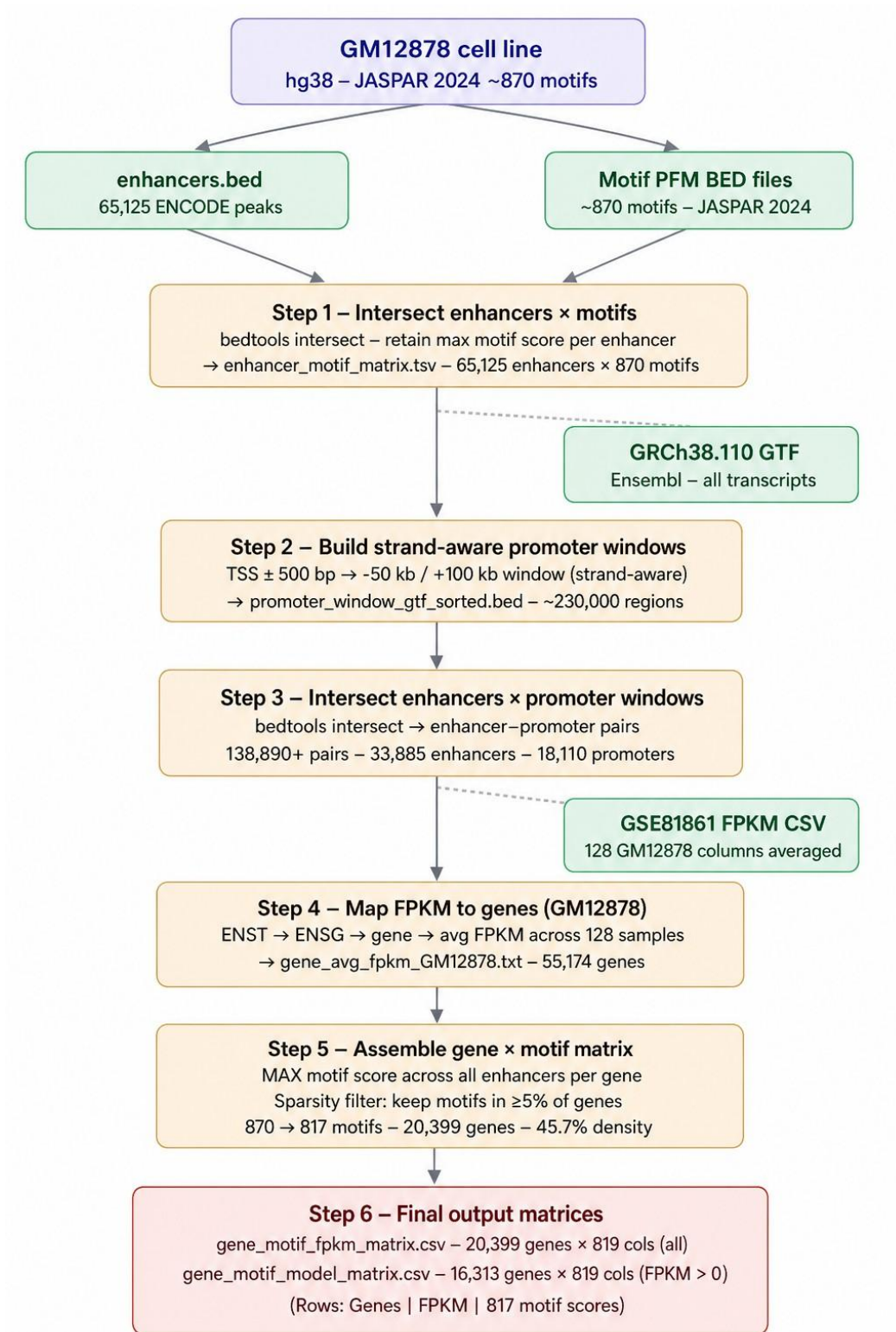


Figure 1. Overview of the gene-motif matrix construction pipeline (150 kb promoter windows).

## 2. Pipeline Steps

### Step 1: Intersect Enhancers with Motif BED Files

Each of the ~870 JASPAR 2024 motif .pfm.bed files was intersected with enhancers.bed using bedtools. For each enhancer, the maximum motif score across all overlapping motif sites was retained.

#### Key command:

```
bedtools intersect -a enhancers.bed -b MOTIF.pfm.bed -wa -wb | awk '{peak=$4; score=$NF; if(score>max[peak]) max[peak]=score} END{for(p in max) print p, max[p]}' > MOTIF.scores.txt
```

#### Output:

- enhancer\_motif\_scores/ — one .scores.txt file per motif
- enhancer\_motif\_matrix.tsv — 65,125 enhancers x 870 motifs (max score or 0.0)

### Step 2: Build Promoter Windows from GTF

Promoter regions were defined as TSS +/- 500 bp from Homo\_sapiens.GRCh38.110.gtf. A strand-aware window of -50 kb to +100 kb was applied around each TSS to capture distal enhancers. GTF-derived promoters were used to guarantee 100% ENST-ENSG coverage (UCSC-sourced promoters matched only 17,192/32,110 ENST IDs).

#### Strand-aware TSS logic:

- Strand '+': win\_start = TSS - 50,000 ; win\_end = TSS + 100,000
- Strand '-': win\_start = TSS - 100,000 ; win\_end = TSS + 50,000

### Step 3: Intersect Enhancers with Promoter Windows

Enhancers were intersected with the strand-aware promoter windows to find which enhancers fall within the regulatory region of each gene.

#### Command:

```
bedtools intersect -a enhancers_sorted.bed -b promoter_window_gtf_sorted.bed -wa -wb > enhancers_in_promoter_windows_gtf.bed
```

| Result                   | Count    |
|--------------------------|----------|
| Enhancer-gene pairs      | 138,890+ |
| Unique enhancers mapped  | 33,885   |
| Unique promoters covered | 18,110   |

#### Step 4: Map FPKM to Genes (GM12878)

Gene expression values (FPKM) were computed from GSE81861\_Cell\_Line\_FPKM.csv by averaging all 128 GM12878 columns (B1 + B2 batches) per gene.

| FPKM Range    | Gene Count  |
|---------------|-------------|
| FPKM = 0      | 4,086 genes |
| FPKM 0 - 10   | 9,541 genes |
| FPKM 10 - 100 | 5,248 genes |
| FPKM > 100    | 1,524 genes |

#### Step 5: Build Gene x Motif Matrix

For each gene, the maximum motif score across all enhancers in its promoter window was taken for each motif, collapsing many enhancer-per-gene rows into a single gene row. A 5% sparsity filter retained 817 of 870 motifs.

| Property         | Value             |
|------------------|-------------------|
| Rows (genes)     | 20,399            |
| Columns (motifs) | 817               |
| Non-zero entries | 7,616,151         |
| Matrix density   | 45.7%             |
| Modelling genes  | 16,313 (FPKM > 0) |

### 3. Experimental Conditions

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Three different gene-motif matrix configurations were tested to assess how the choice of regulatory window affects model performance:

| Condition       | Window / Filter            | Genes  | Motifs | Density |
|-----------------|----------------------------|--------|--------|---------|
| 150 kb window   | -50 kb to +100 kb from TSS | 16,313 | 817    | 45.7%   |
| 2 kb window     | +/- 1 kb around TSS        | 10,785 | 732    | Higher  |
| H2BK20ac filter | Active enhancers only      | 3,021  | 761    | Lower   |

#### 3.1 Condition 1: 150 kb Promoter Window

The primary analysis linked each gene to all enhancers within a 150 kb strand-aware window around its TSS. This is the most inclusive window and captures the majority of known cis-regulatory interactions.

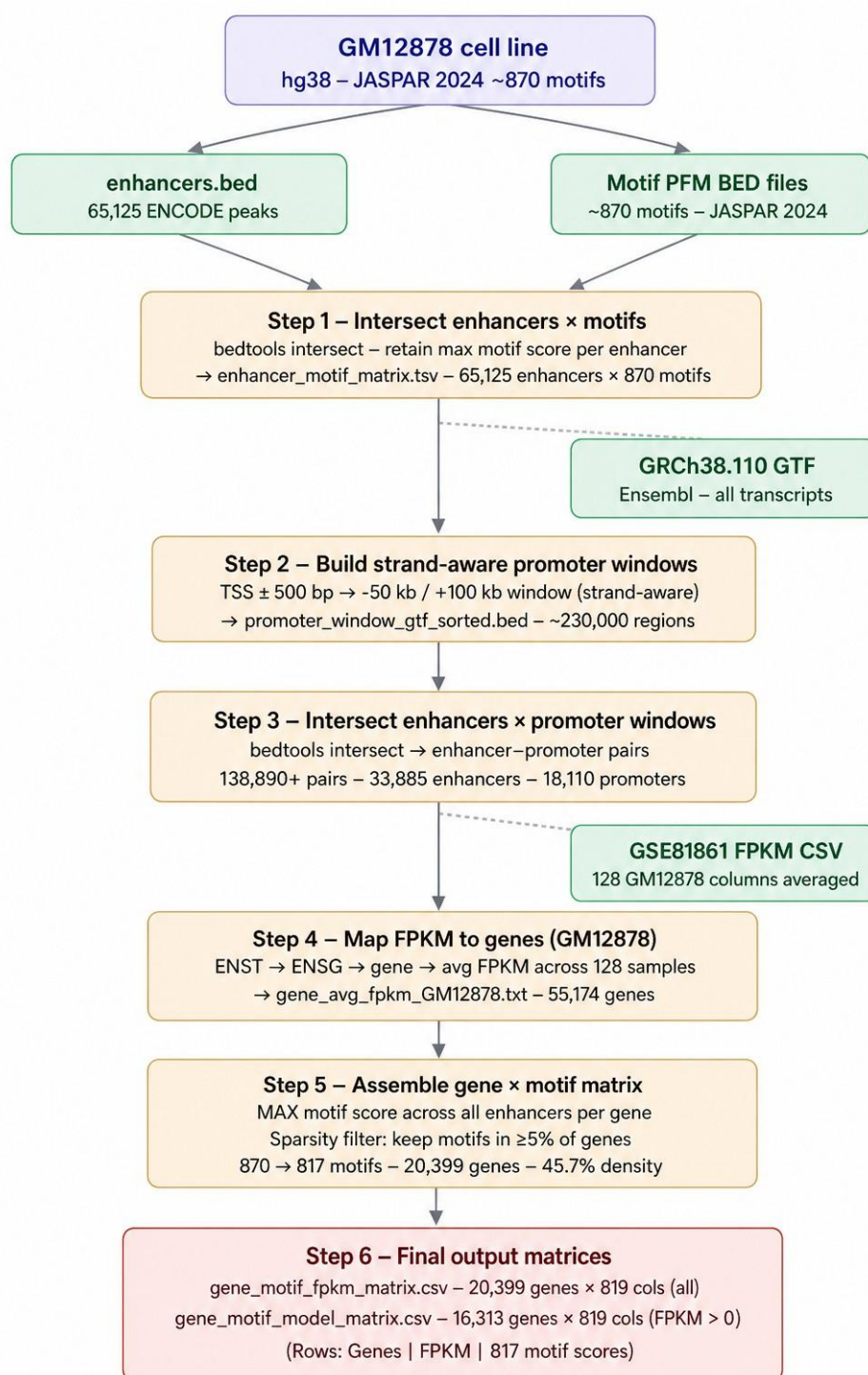


Figure 2. Pipeline diagram for 150 kb promoter window matrix construction.

### 3.2 Condition 2: 2 kb Promoter Window

A narrow 2 kb window ( $\pm 1$  kb around TSS) was tested to evaluate whether proximal promoter motifs alone are sufficient for gene expression prediction. This captures only immediate promoter elements.

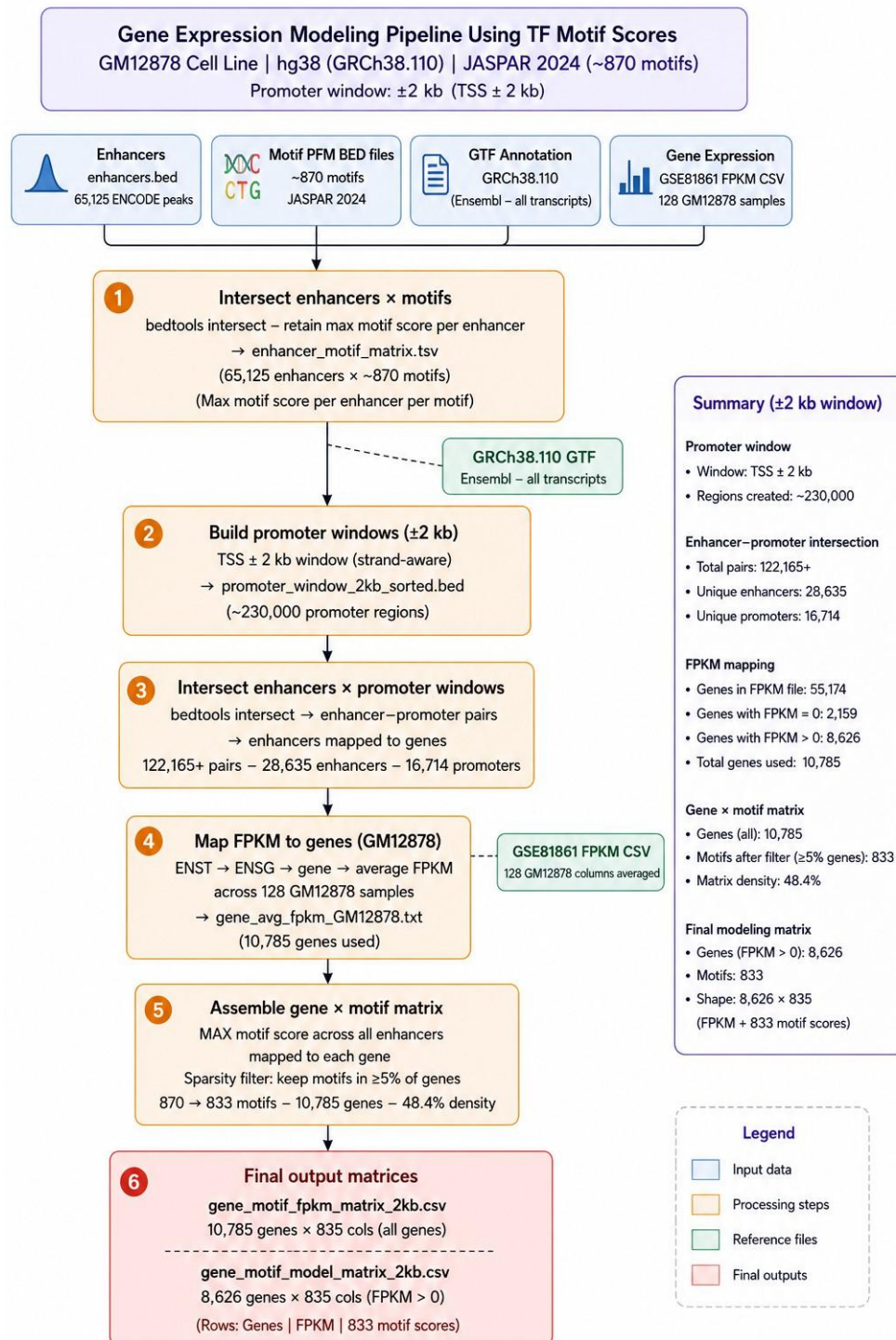


Figure 3. Pipeline diagram for 2 kb promoter window matrix construction.

### 3.3 Condition 3: H2BK20ac Active Enhancer Filter

H2BK20ac (histone H2B lysine 20 acetylation) ChIP-seq peaks from GM12878 were used to filter enhancers to only those that are actively marked in the cell line. This reduces the enhancer set from 65,125 to ~1,878 active peaks, making the model more cell-type specific.

## Bash command used:

```
grep "GM12878" His.Bld.05.H2BK20ac.AllCell.bed | awk '{print $1"\t"$2"\t"$3}' > GM12878_H2BK20ac_3col.bed
```

```
bedtools intersect -a enhancers_sorted.bed -b GM12878_H2BK20ac_sorted.bed -wa -u > active_enhancers_H2BK20ac.bed
```

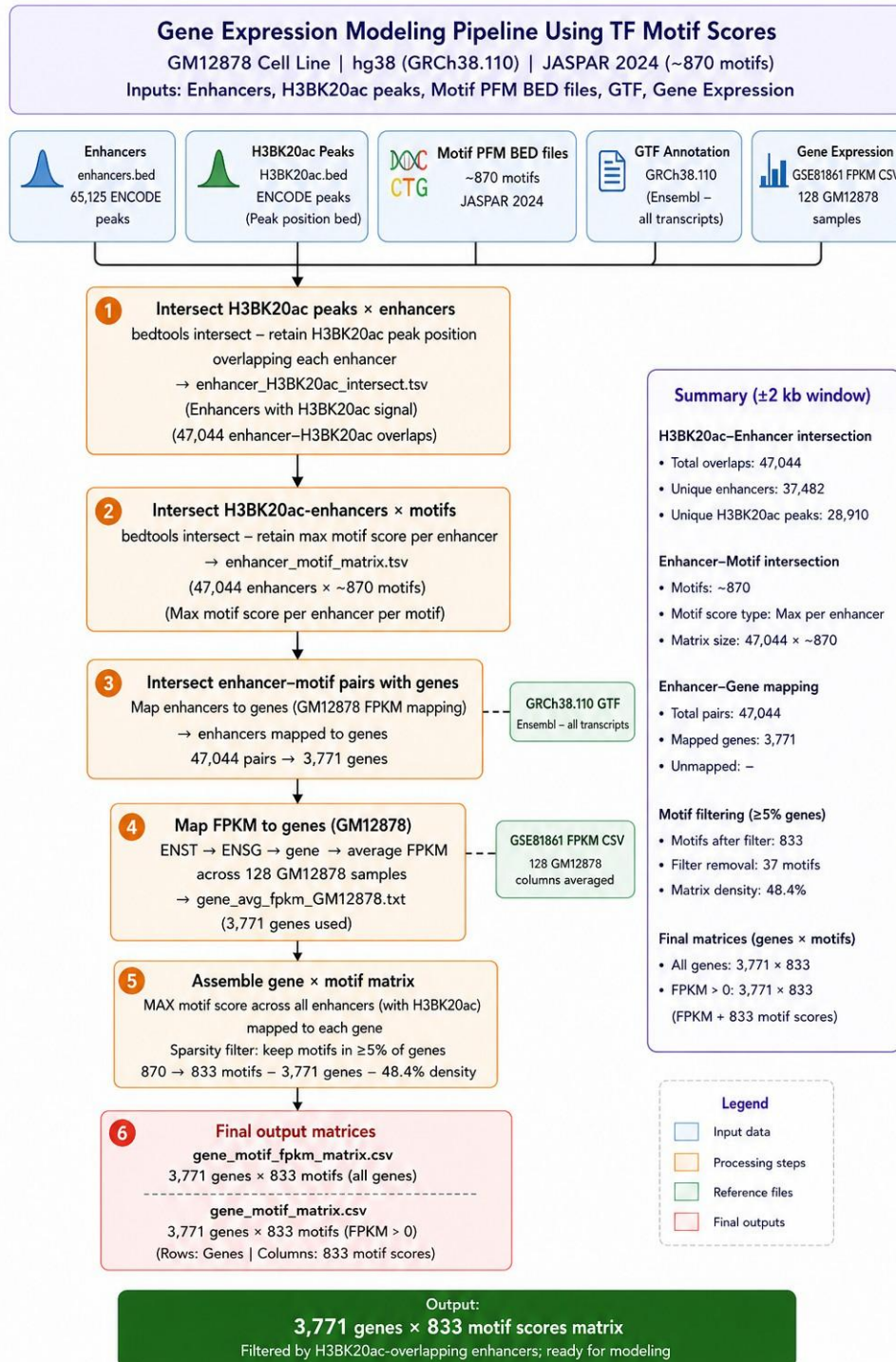


Figure 4. Pipeline diagram for H2BK20ac active enhancer filter.

## 4. Correlation Analysis

Pearson and Spearman correlations were computed between each TF motif score and  $\log^2(1+\text{FPKM})$  to identify motifs most predictive of gene expression. Pearson correlation captures linear relationships while Spearman (rank-based) is more robust to outliers and non-linear associations.

### 4.1 Condition 1: 150 kb Window

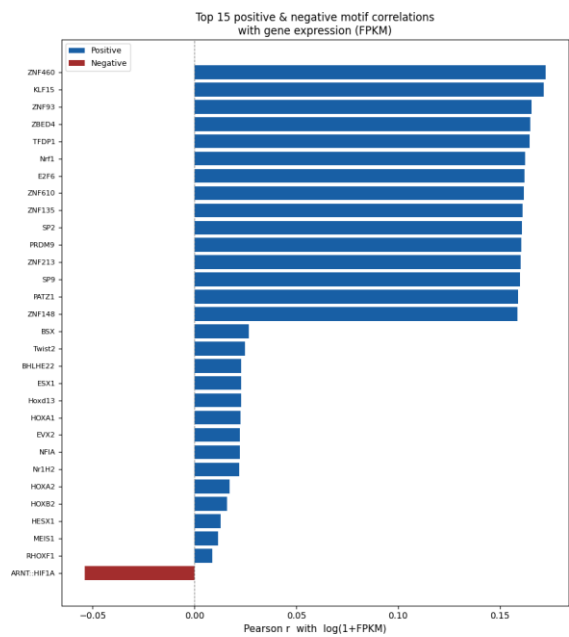


Figure 5a. Top correlated motifs (150 kb window).

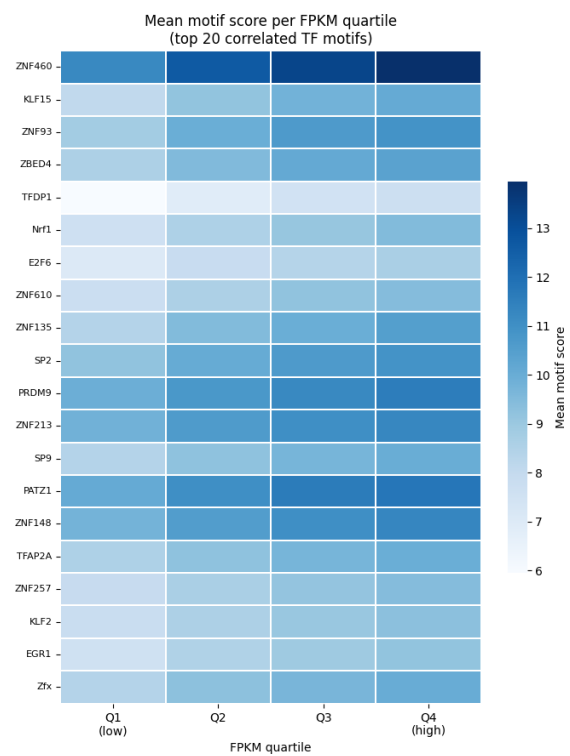


Figure 5b. Correlation distribution (150 kb window).

## 4.2 Condition 2: 2 kb Window

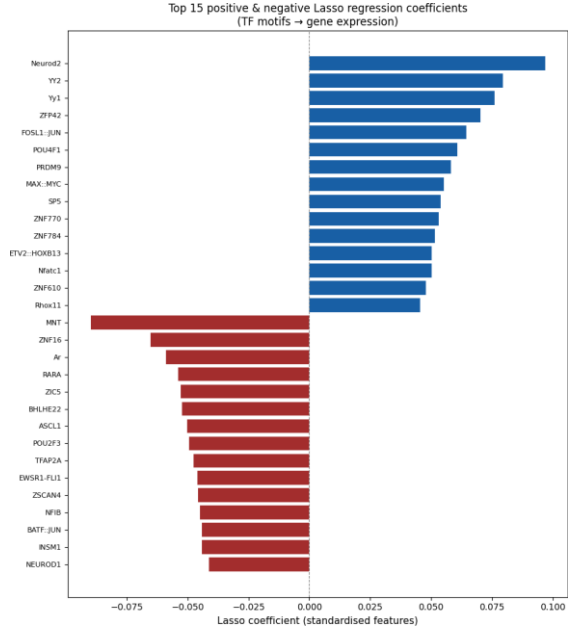


Figure 6a. Top correlated motifs (2 kb window).

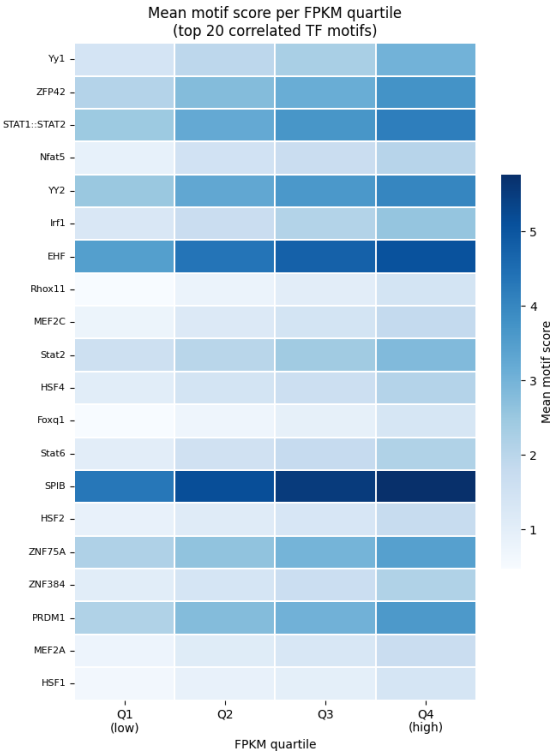


Figure 6b. Correlation distribution (2 kb window).

## 4.3 Condition 3: H2BK20ac Filter

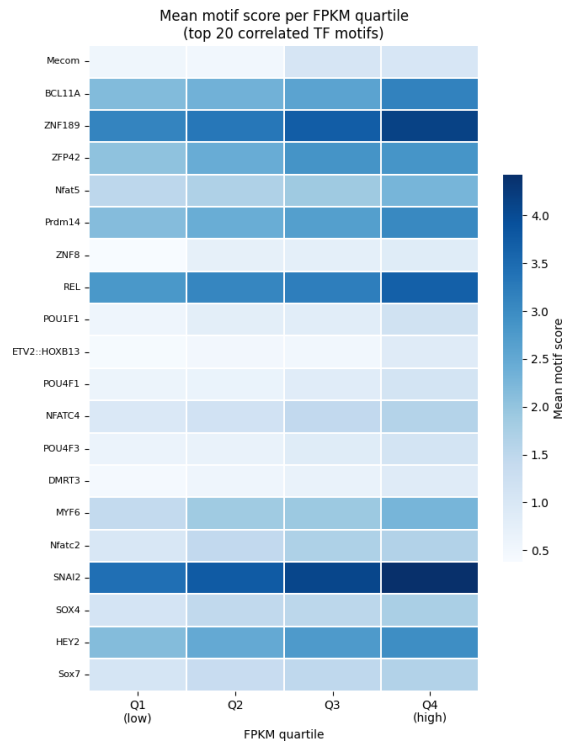


Figure 7a. Top correlated motifs (H2BK20ac filter).

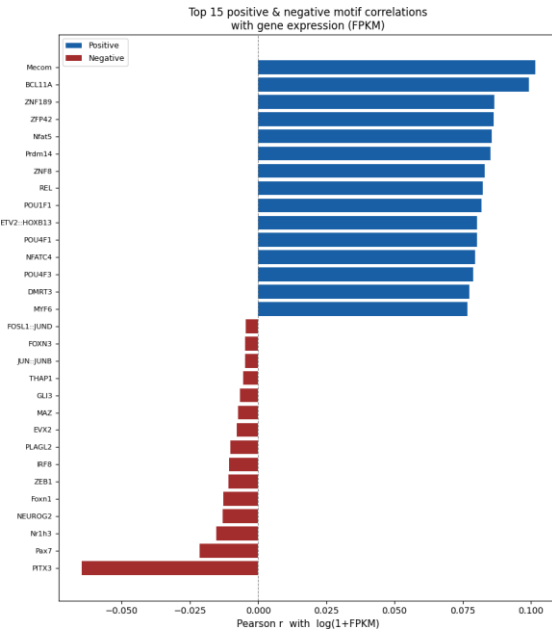


Figure 7b. Correlation distribution (H2BK20ac filter).

## 5. LASSO Regression Modelling

LASSO (Least Absolute Shrinkage and Selection Operator) regression was applied to predict  $\log^2(1+\text{FPKM})$  from motif scores. LASSO applies L1 regularisation which drives uninformative motif coefficients to exactly zero, performing automatic feature selection — particularly valuable given 817+ input features. An 80/20 train/test split was used throughout. LassoCV with 5-fold cross-validation was used to select the optimal regularisation strength (alpha).

### 5.1 Condition 1: 150 kb Window — LASSO Results

| Metric          | Value                     |
|-----------------|---------------------------|
| Dataset shape   | 16,313 genes x 817 motifs |
| Train split     | 13,050 genes (80%)        |
| Test split      | 3,263 genes (20%)         |
| Alpha           | 0.01 (fixed)              |
| Non-zero motifs | 321 / 817                 |
| Train R2        | 0.1240                    |
| Test R2         | 0.1020                    |
| Train Pearson r | 0.3555                    |
| Test Pearson r  | 0.3212 (p = 3.46e-79)     |
| Train RMSE      | 1.6351 (log scale)        |
| Test RMSE       | 1.7002 (log scale)        |
| 5-fold CV R2    | 0.0767 +/- 0.0229         |

```

Training set : 13050 genes
Test set    : 3263 genes
=====
Lasso Regression Results (alpha=0.01)
=====
Train R2      : 0.1240
Test R2      : 0.1020
Train RMSE   : 1.6351 (log scale)
Test RMSE    : 1.7002 (log scale)
Train Pearson r : 0.3555 (p=0.00e+00)
Test Pearson r : 0.3212 (p=3.46e-79)
Non-zero motifs : 321 / 817
=====
5-fold CV R2 scores : [0.0817 0.0746 0.0928 0.0346 0.0997]
Mean CV R2         : 0.0767 ± 0.0228

```

Figure 8a. LASSO summary metrics (150 kb).

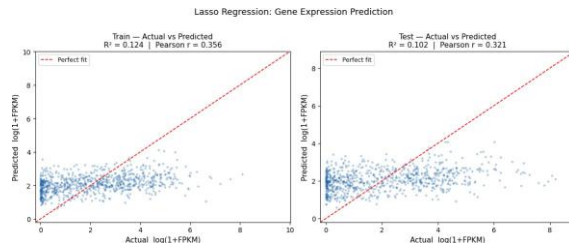


Figure 8b. Actual vs Predicted scatter (150 kb).

## 5.2 Condition 2: 2 kb Window — LASSO Results

| Metric          | Value                     |
|-----------------|---------------------------|
| Dataset shape   | 10,785 genes x 732 motifs |
| Train split     | 8,628 genes (80%)         |
| Test split      | 2,157 genes (20%)         |
| Alpha           | 0.01 (fixed)              |
| Non-zero motifs | 333 / 732                 |
| Train R2        | 0.1840                    |
| Test R2         | 0.1437                    |
| Train Pearson r | 0.4332                    |
| Test Pearson r  | 0.3791 (p = 1.09e-74)     |
| Train RMSE      | 1.5034 (log scale)        |
| Test RMSE       | 1.5167 (log scale)        |
| 5-fold CV R2    | 0.1014 +/- 0.0115         |

```

Training set : 8628 genes
Test set    : 2157 genes
=====
Lasso Regression Results (alpha=0.01)
=====
Train R2      : 0.1840
Test  R2      : 0.1437
Train RMSE     : 1.5034 (log scale)
Test  RMSE     : 1.5167 (log scale)
Train Pearson r : 0.4332 (p=0.00e+00)
Test  Pearson r : 0.3791 (p=1.09e-74)
Non-zero motifs : 333 / 732
=====
5-fold CV R2 scores : [0.1059 0.1117 0.1041 0.0799 0.1056]
Mean CV R2       : 0.1015 ± 0.0111

```

Figure 9a. LASSO summary metrics (2 kb).

### Lasso Regression: Gene Expression Prediction

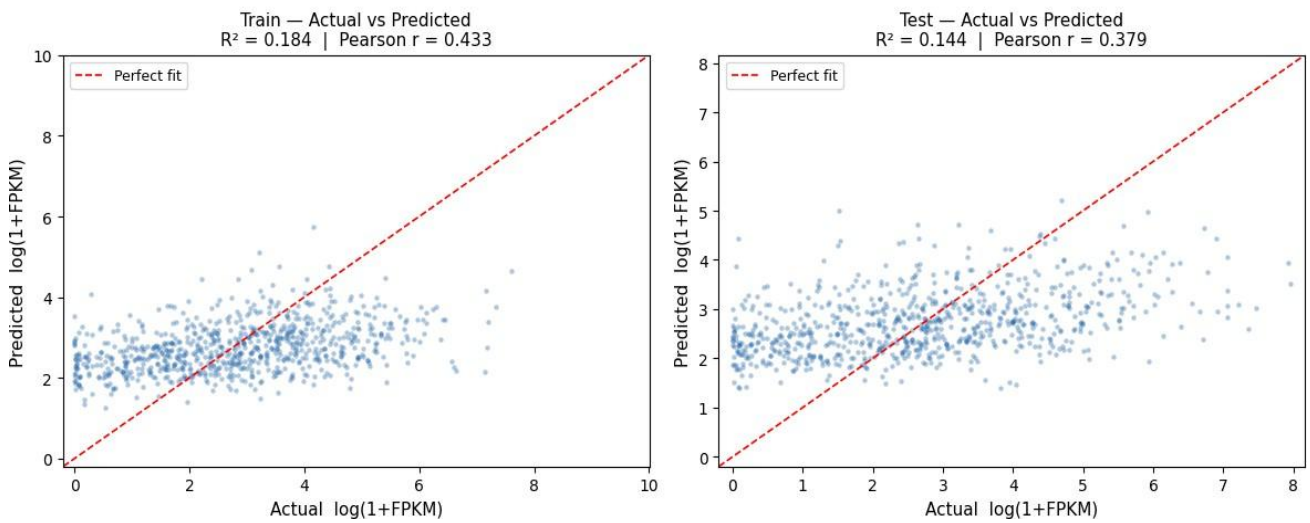


Figure 9b. Actual vs Predicted scatter (2 kb).

5.3 Condition 3: H2BK20ac Filter — LASSO Results (LassoCV)

| Metric          | Value                                        |
|-----------------|----------------------------------------------|
| Dataset shape   | 2,515 genes x 761 motifs (FPKM > 0, clipped) |
| Train split     | 2,263 genes (80%)                            |
| Test split      | 252 genes (20%)                              |
| Best alpha (CV) | 0.060927 (via LassoCV)                       |
| Non-zero motifs | 23 / 177                                     |
| Train R2        | 0.0309                                       |
| Test R2         | 0.0115                                       |

| Metric          | Value                 |
|-----------------|-----------------------|
| Train Pearson r | 0.1921 (p = 2.97e-20) |
| Test Pearson r  | 0.1109 (p = 7.89e-02) |
| Train RMSE      | 1.7358 (log scale)    |
| Test RMSE       | 1.7667 (log scale)    |
| 5-fold CV R2    | 0.0089 +/- 0.0078     |

Lasso Regression: Gene Expression Prediction

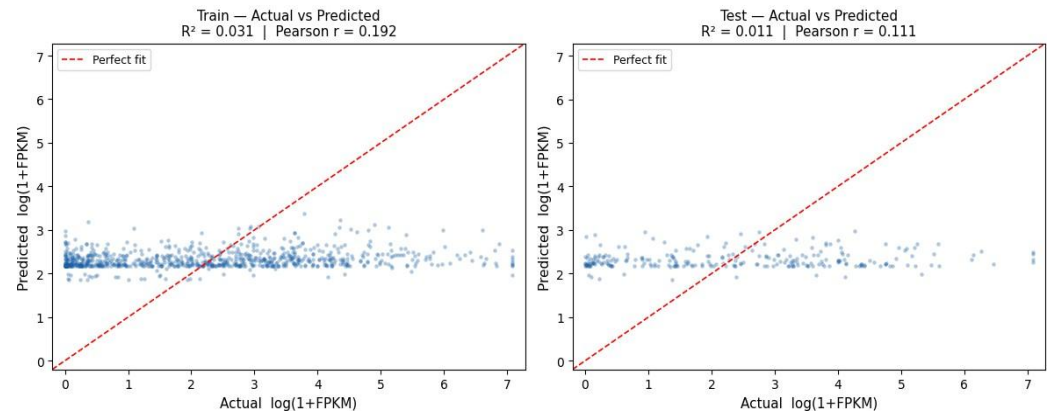


Figure 10b. Actual vs Predicted scatter (H2BK20ac).

```

=====
Lasso Regression Results (alpha=0.06093)
=====
Train R2      : 0.0309
Test  R2      : 0.0115
Train RMSE     : 1.7358 (log scale)
Test  RMSE     : 1.7667 (log scale)
Train Pearson r : 0.1921 (p=2.97e-20)
Test  Pearson r : 0.1109 (p=7.89e-02)
Non-zero motifs : 23 / 177
=====

5-fold CV R2 scores : [0.0008 0.0042 0.0212 0.0033 0.0149]
Mean CV R2       : 0.0089 ± 0.0078

```

Figure 10a. LASSO summary metrics (H2BK20ac)

## 6. Model Comparison Across Conditions

The table below summarises LASSO performance across all three experimental conditions. The 2 kb window achieves the best test R<sup>2</sup> and Pearson correlation, suggesting proximal promoter motifs are more informative per gene than distant enhancers. The H2BK20ac condition suffers from the much smaller gene set after active-enhancer filtering.

| Condition       | Genes  | Motifs | Test R <sup>2</sup> | Test Pearson r | Test RMSE |
|-----------------|--------|--------|---------------------|----------------|-----------|
| 150 kb window   | 16,313 | 817    | 0.1020              | 0.3212         | 1.7002    |
| 2 kb window     | 10,785 | 732    | 0.1437              | 0.3791         | 1.5167    |
| H2BK20ac filter | 2,515  | 761    | 0.0115              | 0.1109         | 1.7667    |

## 7. Advanced Modelling: DNN and Feature Ablation

A combined feature matrix was built by concatenating promoter motifs (2 kb), enhancer motifs (150 kb), and histone/sequence features. Multiple models were evaluated on this combined matrix using an 80/20 stratified split.

### 7.1 Multi-Model Comparison

| Model                | R <sup>2</sup> | Pearson r | Spearman rho | RMSE   |
|----------------------|----------------|-----------|--------------|--------|
| Ridge (all features) | 0.1415         | 0.4556    | 0.4533       | 1.9352 |
| Branched DNN         | 0.1489         | 0.4954    | 0.4830       | N/A    |
| XGBoost (all)        | 0.3188         | 0.5656    | 0.5587       | 1.7238 |

|                                |        |         |          |        |
|--------------------------------|--------|---------|----------|--------|
| === FINAL MODEL COMPARISON === |        |         |          |        |
|                                | R2     | Pearson | Spearman | RMSE   |
| Branched DNN                   | 0.0943 | 0.4914  | 0.4847   | NaN    |
| Ridge (all)                    | 0.1415 | 0.4556  | 0.4533   | 1.9352 |
| XGBoost (all)                  | 0.3188 | 0.5656  | 0.5587   | 1.7238 |

Figure 11. Multi-model comparison results.

## 7.2 Feature Ablation Study

To understand which feature groups drive prediction, an ablation study was conducted by training Ridge regression on each feature subset separately:

| Feature Group             | R2      | Pearson r |
|---------------------------|---------|-----------|
| PROM only (2 kb motifs)   | 0.0372  | 0.2894    |
| ENH only (150 kb motifs)  | -0.0359 | 0.1776    |
| HIST only (histone+seq)   | 0.1770  | 0.4224    |
| PROM + ENH (no histone)   | -0.0156 | 0.2993    |
| PROM + HIST (no enhancer) | 0.1961  | 0.4696    |
| ENH + HIST (no promoter)  | 0.1501  | 0.4263    |
| ALL features              | 0.1415  | 0.4556    |

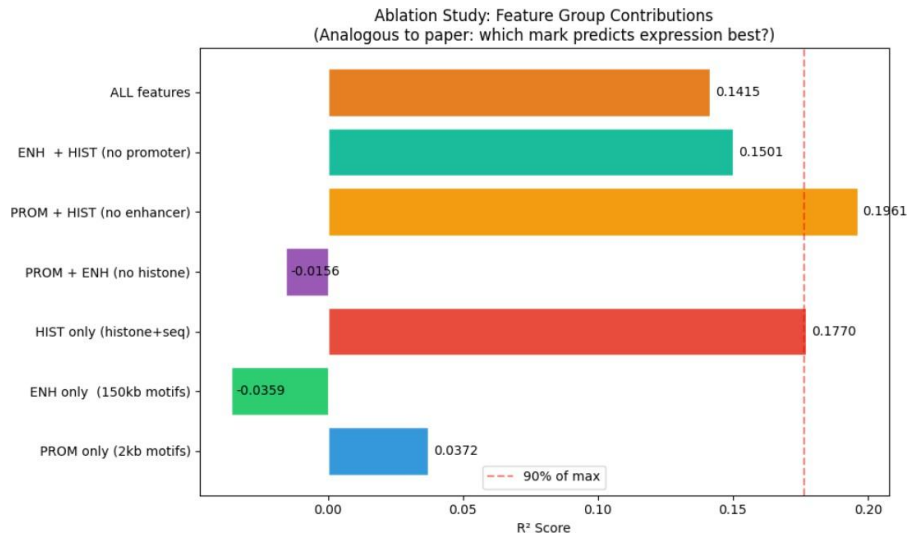


Figure 12. Feature contribution analysis — ablation study results.

## 7.3 Branched DNN Architecture

A branched deep neural network was implemented with three separate input branches (promoter motifs, enhancer motifs, histone/sequence features), each processed independently before concatenation and final regression. The model was trained for 300 epochs with early stopping, dropout (0.3), L2 regularisation (1e-4), and Adam optimiser (lr = 1e-3).

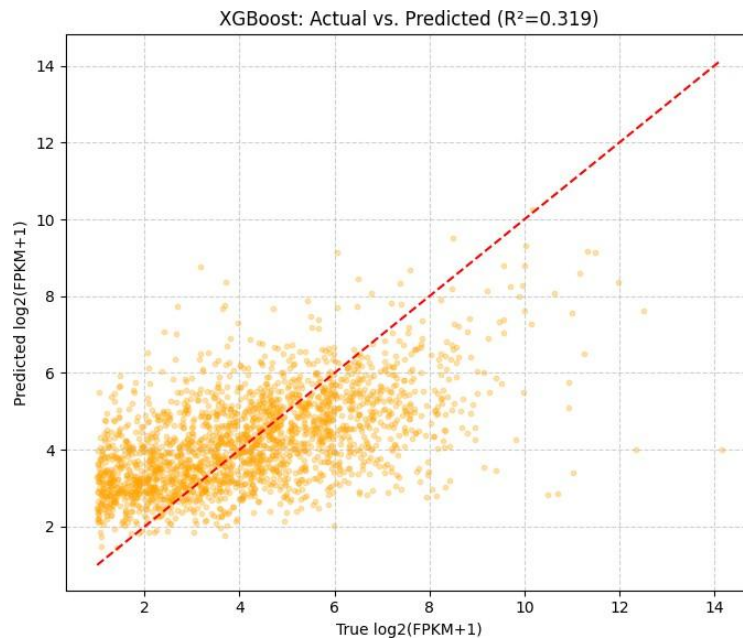


Figure 13. XgBoost actual vs predicted scatter.

## 8. Key Findings and Biological Interpretation

The following insights emerged from the full modelling pipeline:

**XGBoost achieves the best overall performance.**  $R^2 = 0.3188$ , Pearson  $r = 0.5656$  — substantially outperforming linear models, suggesting non-linear interactions between TF motifs contribute to expression.

**Histone features dominate.** The ablation study shows that histone/sequence features (HIST only) give  $R^2 = 0.177$  and Pearson  $r = 0.422$ , outperforming both promoter and enhancer motifs individually. This aligns with chromatin accessibility being a primary determinant of gene expression.

**2 kb window outperforms 150 kb for LASSO.** The narrow 2 kb window gives Test  $R^2 = 0.1437$  vs 0.1020 for 150 kb. Proximal promoter elements appear more consistently predictive than distal enhancers under linear modelling.

**H2BK20ac filtering reduces gene coverage severely.** Restricting to H2BK20ac-marked active enhancers yields only 2,515 usable genes, limiting statistical power. The resulting LASSO model shows near-zero performance (Test  $R^2 = 0.0115$ ).

**Motif selection by LASSO is sparse but consistent.** LASSO selects approximately 320-366 motifs out of 730-817, indicating that roughly 40-45% of JASPAR motifs carry some predictive signal for gene expression in GM12878.

## 9. Important Notes and Limitations

- FPKM values are  $\log^2$ -transformed ( $\log_2(\text{FPKM}+1)$ ) before modelling to reduce skew

from highly expressed genes.

- Motif scores are log-odds scores from JASPAR PFM scanning; higher score = stronger predicted TF binding affinity.
- The 5% sparsity filter retains 817/870 motifs; a lower threshold (e.g. 1%) includes more motifs but increases sparsity.
- One enhancer can be linked to multiple genes if it falls within multiple promoter windows; each gene gets the max motif score across all its enhancers.
- Non-coding RNAs (SNORA, SNORD, LINC, MIR, 7SK etc.) were not explicitly filtered but are largely absent from the FPKM matrix.
- UCSC-sourced promoters (promoters\_clean.bed) were tested but discarded as only 17,192/32,110 ENST IDs matched the Ensembl GTF.
- GM12878-specific FPKM was recomputed directly from GSE81861\_Cell\_Line\_FPKM.csv for 55,174 genes (not gene\_avg\_fpkm\_clean.txt which covers only 15,183).

## 10. Output Files

| File                              | Shape        | Description                                 |
|-----------------------------------|--------------|---------------------------------------------|
| enhancer_motif_matrix.tsv         | 65,125 x 870 | Raw enhancer x motif max scores             |
| gene_motif_fpkm_matrix.csv        | 20,399 x 819 | Gene x motif matrix with FPKM (all genes)   |
| gene_motif_model_matrix.csv       | 16,313 x 819 | Modelling matrix (FPKM > 0 genes only)      |
| gene_motif_model_matrix_2kb.csv   | 10,785 x 734 | Modelling matrix — 2 kb window              |
| gene_motif_H2BK20ac_matrix.csv    | 3,021 x 763  | Modelling matrix — H2BK20ac filter          |
| pearson_spearman_correlations.csv | All motifs   | Pearson + Spearman r for all motifs vs FPKM |
| lasso_selected_motifs.csv         | Varies       | Non-zero Lasso motifs per condition         |

## 11. Datasets and References

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- [1] GM12878 cell line FPKM values: GSE81861  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE33480>
  - [2] Enhancer data (H3K27ac ChIP-seq): ENCODE  
<https://www.encodeproject.org/files/ENCFF763KGD/>
  - [3] H2BK20ac ChIP-seq: ChIP-Atlas His.Bld.05.H2BK20ac.AllCell.bed
  - [4] Genome annotation: Homo\_sapiens.GRCh38.110.gtf (Ensembl release 110)
  - [5] Motif database: JASPAR 2024 <https://jaspar.elixir.no/>
  - [6] Uniform, optimal signal processing of mapped deep-sequencing data. PubMed  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4864461/>
-